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United States Patent	12414654
Kind Code	B2
Date of Patent	September 16, 2025
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Micro puree machine with fixed motors

Abstract

A micro puree machine has a mixing shaft attachable to a blade that is subject to both linear and rotational movement. A drive motor is contained within a lower portion of the machine housing and operatively couples to a first gear that engages a second gear on the mixing shaft for rotation of the mixing blade. A position motor within the upper housing operates to move the mixing blade linearly to engage the ingredients in the bowl. During both axial and rotational movement of the mixing shaft, the drive motor and the position motor remain stationary within their respective positions in the housing.

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Appl. No.: 18/530794

Filed: December 06, 2023

Prior Publication Data

Document Identifier	Publication Date
US 20240164584 A1	May. 23, 2024

Related U.S. Application Data

continuation parent-doc US 17992278 20221122 US 11882965 child-doc US 18530794
continuation parent-doc WO PCT/CN2022/123017 20220930 PENDING child-doc US 17992278

Publication Classification

Int. Cl.: A47J43/08 (20060101); A47J43/07 (20060101); A47J43/044 (20060101)

U.S. Cl.:

CPC A47J43/082 (20130101); A47J43/0711 (20130101); A47J2043/04472 (20130101)

Field of Classification Search

CPC: A47J (43/082); A47J (43/0711); A47J (2043/04472); A47J (43/044); A47J (43/0705); A47J (43/087); A47J (2043/0449)

USPC: 99/510

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of U.S. application Ser. No. 17/992,278, filed on Nov. 22, 2022, entitled MICRO PUREE MACHINE WITH FIXED MOTORS, which in turn is a continuation of International Application No. PCT/CN2022/123017, filed on Sep. 30, 2022, the entire contents of which are incorporated herein by reference for all purposes.

FIELD OF THE INVENTION

(1) The present disclosure relates to a food processing device and, more particularly, to a micro puree machine for making frozen foods and drinks.

BACKGROUND

(2) Home use machines that are intended to make ice creams, gelatos, frozen yogurts, sorbets and the like are known in the art. Typically, a user adds a series of non-frozen ingredients to a bowl. The ingredients are then churned by a paddle while a refrigeration mechanism simultaneously freezes the ingredients. These devices have known shortcomings including, but not limited to, the amount of time and effort required by the user to complete the ice cream making process. Machines of this nature are impractical for preparing most non-dessert food products.

(3) An alternative type of machine known to make a frozen food product is a micro-puree machine. Typically, machines of this nature spin and plunge a blade into a pre-frozen ingredient or combination of ingredients. While able to make frozen desserts like ice creams, gelatos, frozen yogurts, sorbets and the like, micro puree style machines can also prepare non-dessert types of foods such as non-dessert purees and mousses. In addition, consumers can prepare either an entire batch of ingredients or a pre-desired number of servings.

(4) Some current micro puree machines have both a position motor for moving the spinning blade up and down relative to the ingredients, and a separate drive motor for rotating the blade. In these cases, the drive motor may be configured to move up and down with the position motor.

SUMMARY

(5) This disclosure describes a micro puree machine in which the mixing shaft attachable to the blade is subject to both linear and rotational movement. A drive motor is contained within a lower portion of the machine housing and operatively couples to a first gear that engages a second gear on the mixing shaft for rotation of the mixing blade. A position motor within the upper housing operates to move the mixing blade linearly to engage the ingredients in the bowl. During both the axial and rotational movement of the mixing shaft, the drive motor and the position motor remain stationary within their respective positions in the housing. This configuration advantageously makes for a less complicated internal drive mechanisms and/or a simpler, more cost effective and robust machine.

(6) Embodiments of the micro puree machine of this disclosure may include one or more of the following, in any suitable combination.

(7) In embodiments, a micro puree machine of this disclosure includes a transmission system. The transmission system includes a drive motor operatively coupled to a first gear, a mixing shaft operatively coupled to a second gear, and a position motor operatively coupled to the mixing shaft to move the mixing shaft axially along a mixing shaft axis. The first gear is configured to drive the second gear to rotate the mixing shaft about the mixing shaft axis. During axial and rotational movement of the mixing shaft, the drive motor and the position motor are configured to remain in a stationary position relative to a housing of a micro puree machine.

(8) In further embodiments, the drive motor extends along a drive motor axis, and the mixing shaft axis extends in a non-parallel direction relative to the drive motor axis. In embodiments, the mixing shaft axis extends at about a 45 degree to about a 55 degree angle relative to the drive motor axis. In embodiments, the position motor is positioned above the drive motor and/or the housing includes an upper housing, and the position motor is positioned within the upper housing. In embodiments, the housing further includes a base for supporting the upper housing. In

embodiments, the housing further includes a middle housing extending between the base and the upper housing, and the drive motor is positioned with the middle housing. In embodiments, the drive motor is operatively coupled to the first gear via a plurality of planetary gears. In embodiments, the first and second gears are bevel gears. In embodiments, the drive motor is an AC motor and the position motor is a DC motor.

(9) In yet further embodiments, the mixing shaft is coupled to a first end of an inner shaft. The mixing shaft and the inner shaft are axially moveable along the mixing shaft axis within a fixed outer shaft. In embodiments, a second end of the inner shaft is coupled to a carriage within the housing. In embodiments, at least one threaded rod extends through the carriage. In embodiments, the position motor is operatively coupled to a plurality of gears housed within a gearbox, and one of the plurality of gears engages the at least one threaded rod for rotation of the at least one threaded rod relative to the carriage. In embodiments, the carriage is rotationally constrained such that rotation of the at least one threaded rod relative to the carriage causes the carriage to move axially along the at least one threaded rod. In embodiments, the carriage is rotationally constrained by at least one pillar extending through the carriage. In embodiments, the at least one threaded rod extending through the carriage includes at least two threaded rods extending through the carriage. In embodiments, the carriage is rotationally constrained by the at least two threaded rods. In embodiments, the transmission system further includes a first microswitch on the carriage and a second microswitch on the gearbox. When the mixing shaft reaches its furthest linear travel limit, the first microswitch activates the second microswitch to change rotational direction of the position motor.

(10) A reading of the following detailed description and a review of the associated drawings will make apparent the advantages of these and other structures. Both the foregoing general description and the following detailed description serve as an explanation only and do not restrict aspects of the disclosure as claimed.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) Reference to the detailed description, combined with the following figures, will make the disclosure more fully understood, wherein:
- (2) FIG. 1 illustrates a housing and bowl of a micro puree machine according to some embodiments of this disclosure;
- (3) FIGS. 2A and 2B illustrate a portion of a transmission system of the micro puree machine of FIG. 1 according to some embodiments;
- (4) FIGS. 3A-3D illustrate the rotational movement of the transmission system of FIGS. 2A and 2B according to some embodiments;
- (5) FIGS. 4A-4C illustrate the linear movement of the transmission system of FIGS. 2A and 2B according to some embodiments;
- (6) FIGS. 5A and 5B illustrate an alternative version of a transmission system of the micro puree machine of FIG. 1 according to some embodiments;
- (7) FIG. 6 shows a linkage assembly including a single universal joint that transfers rotational motion from a drive shaft of a drive motor to a drive shaft of a mixing device in a non-parallel orientation with respect to the drive shaft of the drive motor;
- (8) FIG. 7 shows a linkage assembly using angled rods that transfer rotational motion from a drive shaft of a drive motor to a drive shaft of a mixing device oriented in a non-parallel orientation with respect to the drive shaft of the motor; and
- (9) FIG. 8 shows a portion of a transmission system of the micro puree machine includes a stationary position motor and a stationary drive motor coupled to a mixing drive shaft via a gear

assembly including non-beveled gears.

DETAILED DESCRIPTION

(10) In the following description, like components have the same reference numerals, regardless of different illustrated embodiments. To illustrate embodiments clearly and concisely, the drawings may not necessarily reflect appropriate scale and may have certain structures shown in somewhat schematic form. The disclosure may describe and/or illustrate structures in one embodiment, and in the same way or in a similar way in one or more other embodiments, and/or combined with or instead of the structures of the other embodiments.

(11) In the specification and claims, for the purposes of describing and defining the invention, the terms “about” and “substantially” represent the inherent degree of uncertainty attributed to any quantitative comparison, value, measurement, or other representation. The terms “about” and “substantially” moreover represent the degree by which a quantitative representation may vary from a stated reference without resulting in a change in the basic function of the subject matter at issue. Open-ended terms, such as “comprise,” “include,” and/or plural forms of each, include the listed parts and can include additional parts not listed, while terms such as “and/or” include one or more of the listed parts and combinations of the listed parts. Use of the terms “top,” “bottom,” “above,” “below” and the like helps only in the clear description of the disclosure and does not limit the structure, positioning and/or operation of the feed chute assembly in any manner.

(12) FIG. 1 shows an isometric view of a micro puree machine **10** according to an illustrative embodiment of the present disclosure. The micro puree machine **10** may include a base **100** for supporting an upper housing **140**. A middle housing **120** may extend between the base **100** and the upper housing **140**. The upper housing **140** may include an interface **142** for receiving user inputs to control the micro puree machine **10** and/or display information. The micro puree machine **10** may also include a removable bowl **350**. The bowl **350** may contain one or more ingredients for processing. For example, a user may attach the bowl **350** to a coupling **500** on an angled surface **144** of the upper housing **140** by rotating the bowl **350** relative to the coupling **500**. In embodiments, the bowl **350** may include features **354** on the bottom surface of the bowl **350** to aid in the fixing of frozen ingredients within the bowl **350**. The features **354** may prevent such ingredients from rotational movement within the bowl **350** during use. In embodiments, the bowl **350** can be manufactured from a disposable material to enhance the convenience of using the micro puree machine **10**. Further, the bowl **350** can be sold as a stand-alone item and can also be prefilled with ingredients to be processed during use of the micro puree machine **10**.

(13) FIGS. 2A and 2B illustrate cross-sectional, front views of a portion of a transmission assembly **200** of the micro puree machine **10** according to some embodiments. As shown in FIG. 2A, the transmission assembly **200** may generally include a position motor **260** positioned within the upper housing **140** and a drive motor **244** (FIG. 3A) positioned within the middle housing **120**. A mixing shaft **252** may couple to an inner shaft **246** attached to a moveable carriage **268**. In embodiments, the mixing shaft **252** and the inner shaft **246** may extend along a mixing shaft axis A that is perpendicular to the angled surface **144** of the upper housing **140** (FIG. 1). As shown in FIG. 2B, the mixing shaft **252** and the inner shaft **246** may be axially moveable along the mixing shaft axis A relative to a fixed outer shaft **238**. When the user assembles the bowl **350** to the coupling **500**, a blade **300** inserted into a lid **400** on the bowl **350** may attach to the mixing shaft **252**. Actuation of the position motor **260** and the drive motor **244** manually or via the interface **142** may cause both rotation of the mixing shaft **252** and the blade **300** about the mixing shaft axis A and positioning of the mixing shaft **252** and the blade **300** along the mixing shaft axis A to engage with ingredients inside the bowl **350**. Thus, the mixing shaft **252**, and therefore the blade **300**, may be controlled at different rotational speeds via the drive motor **244** and moved up and down via the position motor **260** in different patterns and speeds to make different food items.

(14) FIGS. 3A-D illustrate the rotational movement of the transmission system **200** according to some embodiments of the disclosure. As shown in FIG. 3A, the transmission system **200** may

include a drive motor assembly **240** disposed in the middle housing **120**. The drive motor assembly **240** may include a drive motor housing **242** and a drive motor **244**. In embodiments, the drive motor **244** may be an AC motor. The drive motor **244** may extend through the middle housing **120** along a drive motor axis V. The drive motor assembly **240** may further include a fan **245** attached to the drive motor **244**. In embodiments, the mixing shaft axis A may extend in a non-parallel direction relative to the drive motor axis V. For example, the mixing shaft axis A may extend at about a 45 degree to about a 55 degree angle relative to the drive motor axis V. However, the disclosure also contemplates that the mixing shaft axis A and the drive motor axis V are substantially parallel. As shown in FIG. 3B, in embodiments, the drive motor **244** may drive a first gear **234** via a gearbox assembly **220**. In embodiments, the first gear **234** may be a bevel gear. The first gear **234** may be configured to rotate a second gear **236** operatively coupled to the mixing shaft **252**. In embodiments, the second gear **236** may be a bevel gear. The second gear **236** may be keyed to the outer shaft **238** through which the inner shaft **246** and the mixing shaft **252** extend. Thus, rotation of the second gear **236** by the first gear **234** may cause rotation of the mixing shaft **252** about the mixing shaft axis A.

(15) FIG. 3C is a detailed view of the gearbox assembly **220** according to some embodiments. As shown in FIG. 3C, the gearbox assembly **220** may include a gearbox housing **222** containing a plurality of planetary gears **224**. In embodiments, the gearbox housing **222** may include upper and lower portions removably attached together. The plurality of planetary gears **224** may operate to deliver power from the drive motor **244** to the first gear **234** for rotating the first gear **234**. FIG. 3D furthermore illustrates a detailed view of the outer shaft **238** and the inner shaft **246** according to some embodiments. In embodiments, the outer shaft **238** may have an inner hexagonal profile while the inner shaft **246** has a corresponding outer hexagonal profile. However, the disclosure contemplates other suitable profiles of the outer shaft **238** and the inner shaft **246** such that rotation of the outer shaft **238** causes rotation of the inner shaft **246** and therefore the mixing shaft **252**.

(16) FIGS. 4A-C illustrate the linear movement of the transmission system **200** according to some embodiments. As shown in FIG. 4A, the position motor **260** may rotate in a first direction to drive a gearbox **262** housing a plurality of gears **270**. The gear **270** most remote from the position motor **260** may engage at least one threaded rod **264**. The threaded rod **264** may rotate inside a threaded nut **266** housed within the carriage **268**. The carriage **268** may be constrained rotationally by one or more pillars **272** extending through the carriage **268**. Thus, as shown in FIG. 4B, rotation of the threaded rod **264** causes linear movement of the carriage **268** relative to the pillar **272** and the threaded rod **264** while the position motor **260**, the pillar **272** and the threaded rod **264** remain stationary within the upper housing **140**. A rod pitch of the threaded rod **264** may be selected to relate to a linear decent rate of the carriage **268**. The linear movement of the carriage **268** may cause axial movement of the inner shaft **246** and the mixing shaft **252** through the outer shaft **238**. As shown in FIG. 4C, when the carriage **268** reaches the end of its travel limit toward the bowl **350**, a first microswitch **274a** on the carriage **268** activates a second microswitch **274b** on the gearbox **262**, which signals the position motor **260** to change rotation direction and return the blade **300** to its home position within the lid **400**.

(17) As illustrated above, embodiments of the transmission assembly **200** can include one threaded rod **264** and one pillar **272**. However, as shown in FIG. 5A, other embodiments of the transmission assembly **200'** can include two threaded rods **264**, each engaged with a different gear **270** within the gearbox **262**. As shown in FIG. 5B, the threaded rods **264** may rotate inside of threaded nuts **266** housed within the carriage **268**. The carriage **268** may be constrained rotationally by both rods **264**. Thus, rotation of the threaded rods **264** causes linear movement of the carriage **268** relative to the threaded rods **264** while the position motor **260** and the threaded rods **264** remain stationary within the upper housing **140**.

(18) Some embodiments of a micro puree machine as described herein may include one or more transmission elements illustrated in connection with FIGS. 6-8 to transmit power from a motor

assembly (e.g., the drive motor assembly **240**) to a shaft (e.g., the mixing shaft **252**).

(19) FIG. **6** shows a linkage assembly **600** including a single universal joint **602** that transfers rotational motion from a drive shaft of a drive motor, e.g., drive motor **244**, via a first coupling **604** to a drive shaft, e.g., shaft **252**, of a mixing device via a second coupling **606** in a non-parallel orientation with respect to the drive shaft of the drive motor;

(20) FIG. **7** shows a linkage assembly **700** using angled rods **702** that transfer rotational motion from a drive shaft of a drive motor, e.g., drive motor **244**, via a first coupling **704** to a drive shaft, e.g., shaft **252**, of a mixing device via a second coupling **706** where the drive shaft, e.g., shaft **252**, of the mixing device, e.g., a blade assembly, is oriented in a non-parallel orientation with respect to the drive shaft of the motor.

(21) FIG. **8** shows a portion of a transmission system **800** within a housing **120** and/or **140** of a micro puree machine **10** having a stationary position motor **804** and a stationary drive motor **802** coupled to a mixing drive shaft **806** via a gear assembly **808** including non-beveled gears.

(22) While the disclosure particularly shows and describes some embodiments, those skilled in the art will understand that various changes in form and details may exist without departing from the spirit and scope of the present application as defined by the appended claims. The scope of this present application intends to cover such variations. As such, the foregoing description of embodiments of the present application does not intend to limit the full scope conveyed by the appended claims.

Claims

1. A micro puree machine comprising a transmission system, the transmission system comprising: a first gear rotatable about a first axis; a mixing shaft operatively coupled to a second gear, the second gear rotatable about a second axis extending in a non-parallel direction relative to the first axis, the second axis being aligned with a central axis of a processing bowl attached to the micro puree machine; and a position motor operatively coupled to the mixing shaft to move the mixing shaft axially along the second axis, the position motor being aligned with the second axis; wherein the first gear is configured to drive the second gear to rotate the mixing shaft about the second axis; and wherein, during axial and rotational movement of the mixing shaft, the position motor is configured to remain in a stationary position relative to a housing of a micro puree machine.
2. The micro puree machine of claim 1, further comprising a drive motor extending along the first axis.
3. The micro puree machine of claim 2, wherein the position motor is positioned above the drive motor.
4. The micro puree machine of claim 2, wherein the housing includes an upper housing, and the position motor is positioned within the upper housing, and wherein the housing further comprises a base for supporting the upper housing.
5. The micro puree machine of claim 4, wherein the housing further comprises a middle housing extending between the base and the upper housing, and the drive motor is positioned with the middle housing.
6. The micro puree machine of claim 2, wherein the drive motor is operatively coupled to the first gear.
7. The micro puree machine of claim 1, wherein the second axis extends at about a 45-degree to about a 55-degree angle relative to the first axis.
8. The micro puree machine of claim 1, wherein the first gear is a bevel gear.
9. The micro puree machine of claim 1, wherein the second gear is a bevel gear.
10. The micro puree machine of claim 2, wherein the drive motor is an AC motor.
11. The micro puree machine of claim 1, wherein the position motor is a DC motor.
12. The micro puree machine of claim 1, wherein the mixing shaft is coupled to a first end of an

inner shaft, the mixing shaft and the inner shaft being axially moveable along the second axis within a fixed outer shaft.

13. The micro puree machine of claim 12, wherein a second end of the inner shaft is coupled to a carriage within the housing.

14. The micro puree machine of claim 13, wherein at least one threaded rod extends through the carriage.

15. The micro puree machine of claim 14, wherein the position motor is operatively coupled to a plurality of gears housed within a gearbox, and wherein one of the plurality of gears engages the at least one threaded rod for rotation of the at least one threaded rod relative to the carriage.

16. The micro puree machine of claim 15, wherein the carriage is rotationally constrained such that rotation of the at least one threaded rod relative to the carriage causes the carriage to move axially along the at least one threaded rod.

17. The micro puree machine of claim 16, wherein the carriage is rotationally constrained by at least one pillar extending through the carriage.

18. The micro puree machine of claim 15, wherein the at least one threaded rod extending through the carriage includes at least two threaded rods extending through the carriage.

19. The micro puree machine of claim 18, wherein the carriage is rotationally constrained by the at least two threaded rods.

20. The micro puree machine of claim 15, wherein the transmission system further comprises a first microswitch on the carriage and a second microswitch on the gearbox and wherein, when the mixing shaft reaches its furthest linear travel limit, the first microswitch activates the second microswitch to change rotational direction of the position motor.
